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Homogeneous equivalent model coupled with P_1-approximation for dense wire meshes volumetric air receivers

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ABSTRACT

A three dimensional homogeneous equivalent model to analyze the thermal behavior of open volumetric receivers made of staggered stack plain-weave wire meshes is presented. The main boundary conditions, optical and geometrical parameters, assumptions and methodology are described. Then, the model is compared with available literature data in a 3D simulation accounting for the inhomogeneous concentrated solar flux profile of the experiments. A parametric study is conducted to study the thermal behavior of different absorbers submitted to a homogeneous concentrated solar flux and to analyze the influence of the heat transfer coefficient and penetration depth.

1. Introduction

Reliable and sustainable supply of energy will become a requisite for further development of the countries. Renewable energies in general and solar thermal electricity (STE) in particular will play an important role in a future 100% renewable energy mix. STE technologies are presented as technologies with the greatest economical and technical potential for energy production in the European Union, Middle East and North Africa, with a total potential of 632 PWh/year [1]. Central Receiver Systems (CRS) or Solar Tower Technology (STT) appears as a promising option that could both increase overall plant efficiency and reduce LCOE.

In contrast to first generation of CRS technology based on technological developments using cavity or external tube receivers with saturated or superheated steam and molten salts schemes [2], the open volumetric air receiver (OVR) technology appears to be a more complex technology but with the potential to achieve higher plant efficiencies; because on the one hand, the volumetric effect that is characteristic of OVR (which means that the absorber temperature at the irradiated side is lower than the outlet air temperature) reduces the frontal heat losses and, on the other hand, the overall plant efficiency is enhanced because the steam turbine can be operated with higher temperatures.

Due to the unique properties of volumetric absorbers: highly porous, large specific surface area and with a highly tortuous and winding flow path, they are receiving a renovated research interest with lot of contributions to this field of knowledge [3–7].

However, since high efficiencies in the thermodynamic cycles are related to high heat transfer fluid (HTF) working temperatures, the most investigated option is the ceramic materials [8–12]. Several studies have been done to predict the thermal behavior of ceramic porous media. Pitz-Paal et al. [13] analyzed the thermal performance and flow instability of a OVR with a three-dimensional irradiance distribution. Wu et al. [14] analyzed in a macroscopic CFD model several geometric and operating parameters using a local thermal non-equilibrium (LTNE) model coupled to the P_1-approximation. Fend et al. [15] modeled both a single-channel and

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a cup of a honeycomb absorber, showing that the finer geometry performed better. Kribus et al. [16] investigated in an uni-dimensional (1D) model the performance of OVR as function of geometrical and material properties giving emphasis on the implementation of the radiative heat transfer. Wang et al. [17] studied the effects of two irradiative transfer models (P_1 and Rosseland approximations) on the porous media with LTNE model. Chen et al. [18,19] studied numerically in a two-dimensional axisymmetric model coupling P_1-approximation to a LTNE model the effect of composite porous structure and double-layer ceramic foam over the OVR performance. Zaversky et al. [20] presented two 1D models with different characteristics using both LTNE validated against experimental data and performing parametric optimization over geometrical foam parameters. Large experimental and numerical work has been performed over the OVR performance, mainly focused on ceramic materials.

On the other hand, metallic materials have not received such interest, despite their advantages; the opportunity to easily work with different configurations and geometries, lightweight structures, lower working temperatures (<800 °C) than ceramic materials that result in lower thermal losses, among others. Recently, Livshits et al. [21] proposed volumetric structures made of dense metallic plain-weave wire mesh screens as an interesting candidate to reach absorber efficiencies exceeding 90% and, Avila-Marin et al. [22,23] numerically studied the convective HTC for six commercially available mesh types and different stack patterns, and the correlations were validated against experimental and numerical results.

This study presents a numerical model and simulates the performance of OVR made with dense metallic wire mesh screens and compares to experimental data of the literature [24–26]. The aim is to numerically analyze the flow, the heat transfer and the temperature profiles of the fluid and solid matrix by solving the coupled volume-averaged governing equations. The radiative heat transfer equation is computed with the P_1-approximation and the LTNE model is used to calculate the temperature profiles.

The article structure is as follows: In section 2, the solar absorber is described; in section 3, the numerical model is presented; in section 4, a comparison with experimental results is given and a parametric study is conducted to analyze the influence of the mesh geometry on the absorber performances. Finally, the conclusions are presented.

2. Solar absorber description

The solar absorber studied in this work consists of a cylindrical pipe that contains different dense wire mesh stacked. The absorbers that are going to be numerically analyzed are those experimentally studied in Avila-Marin [24–26]. For that reason, a three-dimensional model is considered in this work, to account for the non-homogeneous incident concentrated flux used experimentally. Fig. 1 shows the scheme of the solar absorber investigated in this study.

In the experimental work, 6 alloy-310 commercial square plain-weave wire mesh screens were selected with a sample diameter of 50 mm. The unit cell of a plain-weave screen is described by the mesh properties by means of the wire diameter, d, and the mesh size, M. Table 1 presents the mesh properties for 6 alloy-310 screens adopted for the numerical work performed in this paper. The meshes were chosen with the constraint of having pairs with similar porosities and different geometrical (see section 3.3.2 and Table 3) and optical parameters. The thickness of the single porosity absorbers analyzed in this paper were selected experimentally according to a thickness of the single porosity absorber that ensure the total extinction of the incident flux profile in the whole absorber volume [24]. Table 1 shows the number of screens used to construct the 6 volumetric absorbers with single porosity.

Concerning the absorber, there are two main arrangements when plain-weave screens are used for volumetric absorber applications: the inline arrangement, where successive wire mesh screens are aligned and, the stagger arrangement, where successive wire mesh screens are offset in two directions by $0.5 \cdot M^{-1}$. As done experimentally, the staggered stack is adopted in this work.

The front surface of the absorber is subjected to concentrated solar radiation, which is gradually absorbed into the dense wire mesh stack causing a temperature rise. However, if a strong extinction of the incident flux happens in the first millimeters, the wires temperature may reach a high temperature, with possible overheating. The performance of the solar absorber is highly dependent on its porosity, specific surface area and extinction coefficient. The lateral walls of the absorber are well insulated.

3. Mathematical model

The numerical model of the combined flow and heat transfer in

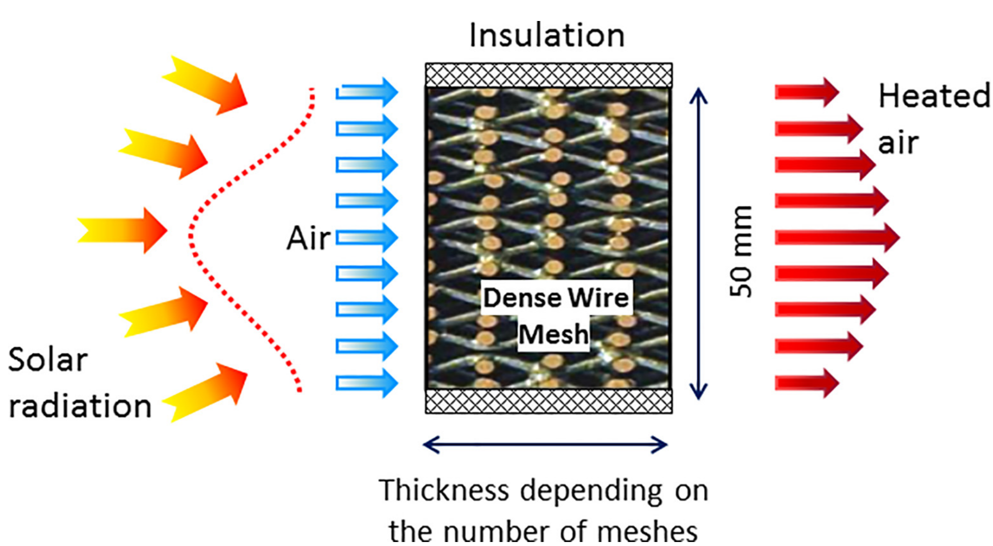


Fig. 1. Schematic diagram of the porous absorber with dense wire mesh.

Table 1
Commercial AISI 310 square plain-weave wire mesh characteristics and number of meshes for single porosity absorbers.

Mesh type			A	B	C	D	E	F
Mesh properties	d	mm	1.00	0.70	0.50	0.16	0.63	0.13
	M	mm^{-1}	0.20	0.31	0.53	1.79	0.61	3.03
Number of meshes	Nº	–	10	9	8	8	5	6

the solar receiver is based on several assumptions: (a) the fluid (air) is Newtonian and behaves like an ideal gas; (b) the fluid flow is at steady state and laminar, because the maximum Reynolds number (based on the hydraulic diameter and superficial velocity), was lower than 200; (c) the physical properties of the fluid phase are temperature dependent; (d) the physical properties of the dense wire mesh stack are temperature dependent; (e) the dense wire mesh is considered gray, absorbing, emitting and scattering is isotropic; (f) lateral walls of solar absorber are adiabatic.

3.1. Governing equations

3.1.1. Mass conservation equation

$$\nabla\cdot\left(\rho_f\cdot\overrightarrow{v_D}\right)=0 \tag{1}$$

Where ρ_f is the fluid density and $\overrightarrow{v_D}$ is the superficial velocity.

3.1.2. Momentum equation

The flow in porous media can be described by the Brinkman-Forchheimer extended Darcy model [12]:

$$\frac{\rho_f}{\varnothing}\langle(v_D\cdot\nabla)\rangle\, v_D=-\nabla\left\langle P_f\right\rangle^f+\frac{\mu_f}{\varnothing}\nabla^2\langle v_D\rangle-\frac{\mu_f}{K_1}\langle v_D\rangle - \frac{\varnothing\cdot\rho_f}{K_2}[|\langle v_D\rangle|\cdot\langle v_D\rangle] \tag{2}$$

Where μ_f is the fluid dynamic viscosity, $\varnothing$ is the porosity, P_f is the fluid pressure, K_1 is the inertial permeability coefficient, K_2 is the viscous permeability coefficient and v_D is the Darcy velocity.

3.1.3. Energy equations

This model considers the local thermal non-equilibrium (LTNE) between both phases, which accounts a difference between the air and solid matrix temperatures. The two equations are linked by the convective source term, using the effective property called the volumetric convective Heat Transfer Coefficient (HTC), h_v.

For the fluid phase:

$$\left(\rho_f\, c_f\right)\langle v_D\rangle\cdot\nabla\left\langle T_f\right\rangle=\nabla\left(k_{eff,f}\cdot\nabla\left\langle T_f\right\rangle\right)+h_v\left(\langle T_s\rangle-\left\langle T_f\right\rangle\right) \tag{3}$$

For the solid phase:

$$0=\nabla\cdot\left(k_{eff,s}\cdot\nabla\langle T_s\rangle-q_r\right)+h_v\cdot\left(\left\langle T_f\right\rangle-\langle T_s\rangle\right) \tag{4}$$

Among the two equations, T_f and T_s represents the temperature of the fluid and the solid phases and, $k_{eff,f}$ and $k_{eff,s}$ mean the effective thermal conductivity of the fluid and the solid phases respectively. h_v is the volumetric convective HTC. q_r is the radiative heat flux source, ρ_f, c_f mean the density and heat capacity of the fluid and the symbols $\langle\rangle$ mean the volumetric average.

The radiative heat flux, q_r, in a gray media could be approximated by:

$$q_r=-\frac{1}{3\cdot(a+\sigma_s)}\cdot\nabla G \tag{5}$$

Where a $(1/\mathrm{m})$ is the absorption coefficient, σ_s $(1/\mathrm{m})$ is the scattering coefficient, and G $(\mathrm{W/m^2})$ is the incident irradiation.

3.1.4. Radiative heat transfer

The radiation inside a porous media working as a solar absorber is a crucial heat transfer mechanisms, and the solution of the radiative transfer equation (RTE) is needed. The RTE describes the radiative intensity field within the geometry as a function of location, direction and spectral variables. The net radiative heat flux crossing a surface element is the sum of the contributions of radiative energy irradiating the surface from all possible directions and for all possible wavenumbers [27], as a result, the solution of the RTE in a coupled heat transfer problem through a porous media requires simplification in order to reduce computation, as the time require can become prohibitive.

Most of CFD studies carried out use the simplification for optically thick media, mainly the Rosseland [28–30] and the P_1-approximations [31–33]. The optically thick approximation considers that the mean free path of photons in the porous media is rather short compared to the mean beam length. As a consequence, the radiation intensity is only affected by the radiation heat transfer in the near region and the radiative intensity tends to be isotropic. This assumption could fail at the boundary.

The present study uses the P_1-approximation to solve the RTE because of its efficient computer time capability and acceptable accuracy for optically thick porous media. The dense mesh screens working as a solar absorber was irradiated with a solar simulator which was assumed to be a collimated incident radiation beam. The P_1-model is used with the intensity splitting technique which uses collimated and diffuse intensities [27]. Thus the incident irradiation computed by the P_1-model is the diffuse irradiation, G_s:

$$-\nabla\left(\frac{1}{3\cdot(a+\sigma_s)}\cdot\nabla G_s\right)=a\cdot\left(4\cdot\sigma\cdot\langle T_s\rangle^4-G_s\right)+\sigma_s\cdot I_0\cdot e^{-\beta\cdot z} \tag{6}$$

Where β is the extinction coefficient, a is the absorption coefficient, σ is the Stefan-Boltzmann constant and, z is the porous absorber thickness. Once the diffuse irradiation, G_s, is solved, the incident irradiation, G, is computed:

$$G=G_c+G_s=I_0\cdot e^{-\beta\cdot z}+G_s \tag{7}$$

And finally the divergence of the radiative heat flux, ∇q_r, is computed as follows:

$$\nabla q_r=a\cdot\left(4\cdot\sigma\cdot\langle T_s\rangle^4-G\right) \tag{8}$$

3.2. Boundary conditions

For all the simulations carried out the following boundary conditions applies:

3.2.1. Concentrated solar incident flux

The incoming concentrated solar incident flux has been set in two different ways (see section 3.3.6):

- Experiment-model comparison: For the model, a user-defined-function was implemented with one representative experimental solar flux profile obtained of the solar simulator assuming collimated radiative flux. However, the radiative flux is actually not parallel, but there was a good agreement between the model results and the experimental data.
- Parametric studies: For the parametric studies the incoming flux is treated as parallel, constant and uniform with a value of 600 kW/m^2.

3.2.2. Inlet conditions

- Fluid: The fluid velocity is set at different values for the parametric study and to that values that match the mass flow rate tested experimentally. The static temperature is fixed to 300 K.
- Solid: The temperature gradient at the solid outlet surface is set to zero.
- P_1-approximation: when the concentrated solar irradiance is split, the Marshak's boundary condition is used at the frontal surface.

$$\frac{-1}{3\cdot\beta}\cdot\frac{dG_{s,in}}{dx}=I_0+2\cdot\sigma\cdot T_{amb}^4-\frac{G_{s,in}(0)}{2} \tag{9}$$

Where $G_{s,in}$ is the diffuse irradiation at the inlet surface, T_{amb} is the ambient temperature, and I_0 is the incident intensity at the entrance.

3.2.3. Outlet conditions

- Fluid: The temperature gradient at the fluid outlet surface and the static pressure (with a reference pressure of 101325 Pa) are set to zero.
- Solid: The temperature gradient at the solid outlet surface is set to zero.
- P_1-approximation: The incident irradiation gradient at the outlet surface is set to zero.

$$\frac{-1}{3\cdot\beta}\cdot\frac{dG_{s,out}}{dx}=0 \tag{10}$$

3.3. Coefficients for the homogeneous equivalent model

The HEM solves macroscopic equations and avoids using the detailed geometry of the porous media, reducing computational time with an acceptable level of accuracy that should be validated with experimental data. The main coefficients needed for the solution of the HEM are shown in Table 2.

3.3.1. Viscous and inertial permeability coefficients

In the momentum equation in Eq. (2) for the HEM, the coefficients K_1 and K_2 refers to the viscous permeability coefficient and the inertial permeability coefficient respectively. Those coefficients were experimentally measured using the apparatus shown in Fig. 2 for the configurations adopted in this study. Some of the measurements are presented in Fig. 3 together with the fitting curve. More information is found in Avila-Marin doctoral thesis [25].

Table 2
Coefficients needed for the homogenous equivalent model.

Equation	Variable	Description	Units
Momentum	K_1	Viscous permeability coefficient	m^2
	K_2	Inertial permeability coefficient	m
	$\varnothing$	Porosity	–
Energy	h_{lv}	Local volumetric heat transfer coefficient	$W/(m^3\cdot K)$
	$k_{eff,f}$	Fluid effective conductivity	$W/(m\cdot K)$
	$k_{eff,s}$	Solid effective conductivity	$W/(m\cdot K)$
Radiative	a	Absorption coefficient	m^{-1}
	σ_s	Scattering coefficient	m^{-1}
	β	Extinction coefficient	m^{-1}
	I_0	Incident radiation	W/m^2

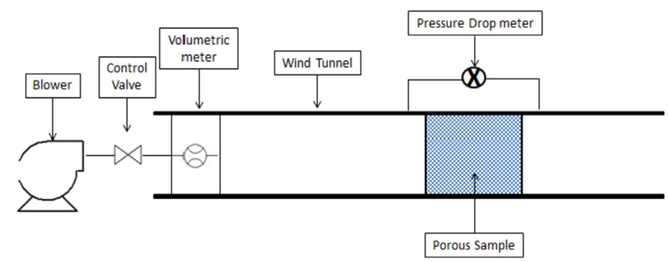


Fig. 2. Experimental setup for the pressure drop measurement.

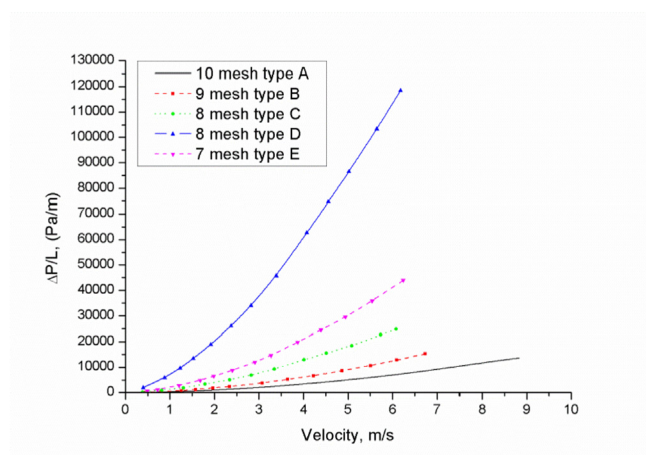


Fig. 3. Pressure drop measurement versus velocity of air for dense wire mesh volumetric absorber.

3.3.2. Geometrical properties

The geometrical parameters that characterized the OVR are essential for the calculation of other related coefficients. This section presents the porosity, the specific surface area, and the mesh hydraulic diameter for plain-weave wire mesh screens.

The porosity, $\varnothing$, of a porous structure is defined as the ratio of the void volume to the total volume of an average representative cell. For a square plain-weave wire mesh screen the non-dimensional parameter C is defined as a function of the geometrical mesh parameters [34] presented in Table 1.

$$C=\left(\frac{M\cdot d}{2}+\frac{1}{2\cdot M\cdot d}\right)\cdot\arcsin\left(\frac{1}{\left(\frac{M\cdot d}{2}+\frac{1}{2\cdot M\cdot d}\right)}\right) \tag{11}$$

So the porosity of a square plain-weave wire mesh screen as a function of the geometric parameters is [34]:

$$\varnothing = 1 - \frac{\pi \cdot M \cdot d \cdot C}{4 \cdot C_f} \tag{12}$$

The constant C_f is the compactness factor and it is used to describe the thickness resulting from different types of packing pattern. When the wire mesh screens follow an inline pattern, the compactness factor equals one; when the wire mesh screens follows a staggered pattern, the compactness factor is less than one.

The specific surface area, a_v, of a porous matrix is defined as the ratio of the total surface area to the total volume. Satisfying the criteria for square screens and using variable C presented previously, the specific surface area for a square plain-weave wire mesh screen is [34]:

$$a_v = \frac{\pi \cdot M \cdot C}{C_f} \tag{13}$$

Finally, it is important to select appropriate characteristic length for the porous media. Among the different length-scale parameters available in the literature, the mesh hydraulic diameter is adopted in this study [34–36]. The hydraulic diameter of a porous media is defined by Kaviany [37]:

$$d_h = \frac{4 \cdot \varnothing}{a_v} \tag{14}$$

Table 3 presents the geometrical properties with stagger stack arrangement, for the single porosity absorbers analyzed in this paper.

3.3.3. Local volumetric heat transfer coefficient

This work considers LTNE model to take into account the large temperature gradient at the entrance of the absorber between the solid matrix and the inlet air. The issue when using the LTNE model is to precisely selects the local volumetric HTC. A large work, either experimental or numerical, has been performed for the determination of the convective HTC [38–41].

The numerical correlations obtained by Avila-Marin [22,23] are being implemented in the HEM for the HTC with LTNE approach. Table 4 presents the coefficients that fit equation (15) for six different wire mesh screens with stagger arrangement. The mesh hydraulic diameter is the characteristic length adopted in these correlations (see Table 3).

$$Nu_{lv} = a \cdot Re^b \cdot Pr^c \tag{15}$$

3.3.4. Effective conductivity

Porous media approach use the effective conductivity for both, the solid and the fluid phases, and treat the conduction as an isotropic media. The effective conductivity of both phases is a mixture of the conductivities of the clear fluid and the dense material. The formulas proposed by Schuetz & Glicksman [42] has been successfully applied by Xia et al. [38], Chen et al. [18,19,32], Meng et al. [28], Wu et al. [14,33], and Kamiuto et al. [43]. The formulas are:

Table 4
Parameters for equation (16) for six dense wire mesh screens with stagger arrangement.

Mesh	*a*	*b*	*c*
A	110.9	0.494	9.19
B	87.7	0.459	8.12
C	69.9	0.352	6.50
D	99.4	0.312	6.84
E	136.9	0.427	9.35
F	1359.0	0.587	16.22

$$k_{eff,f} = \varnothing \cdot k_f \tag{16}$$

$$k_{eff,s} = \frac{1}{3} \cdot (1 - \varnothing) \cdot k_s \tag{17}$$

Where, $k_{eff,f}$ is the effective conductivity of the fluid phase, $k_{eff,s}$ is the effective conductivity of the solid phase, k_f is the fluid conductivity, k_s is the solid conductivity.

3.3.5. Radiative coefficients

The optical properties of the solid matrix are essential for the solution of the RTE. In this work, the optical properties were obtained using the geometrical optics approximation presented by Vafai [44]. The geometrical optics is an approximation that considers the main geometrical shape of the porous media for the evaluation of the optical properties. In the case of foams, the main shape is the sphere and the equations have been widely used and validated in the literature [45–48]. In the case of stacked wire meshes, the main shape of the porous media is a cylinder, and the equations that hold are equations (18)–(20). These equations have been validated in the literature [23] as well. The coefficients are given by the following expressions:

$$a = \frac{4}{\pi} \cdot \alpha \cdot \frac{(1 - \varnothing)}{d} \tag{18}$$

$$\sigma_s = \frac{4}{\pi} \cdot (2 - \alpha) \cdot \frac{(1 - \varnothing)}{d} \tag{19}$$

$$\beta = a + \sigma_s = \frac{8}{\pi} \cdot \frac{(1 - \varnothing)}{d} \tag{20}$$

Where a is the absorption coefficient, σ_s is the scattering coefficient, β is the extinction coefficient, α is the material absorptivity, d is the wire diameter of the mesh screen.

3.3.6. Incident radiation

In this work two different incident radiation profiles are adopted. First, for the experiment-model comparison of the HEM, a representative incident flux profile, measured experimentally from the solar simulator with a Gardon radiometer [24], is implemented

Table 3
Geometrical properties with stagger stack arrangement.

Mesh type			A	B	C	D	E	F
Stagger stack	d	mm	1.00	0.70	0.50	0.16	0.63	0.13
	$\varnothing$	%	70	68	62	62	48	47
	a_v	m^{-1}	1194	1849	3044	9552	3322	16330
	d_h	mm	2.349	1.464	0.814	0.259	0.574	0.115

in the software STAR-CCM+. Then, for the parametric analysis, the incident solar flux is a constant incident flux of 600 kW/m^2.

3.3.7. Air physical properties

Because of the high working temperature of the solar absorber, the air properties are defined by the following polynomial functions [49], with the similar performance to that used by Wu [14]:

The dynamic viscosity is:

$$\mu_{air} = -8.383\cdot 10^{-7} + 8.357\cdot 10^{-8}\cdot T - 7.694\cdot 10^{-11}\cdot T^2 + 4.644\cdot 10^{-14}\cdot T^3 - 1.066\cdot 10^{-17}\cdot T^4 \quad (21)$$

The thermal capacity at constant pressure is:

$$c_{air} = 1047.637 - 0.3726\cdot T + 0.000945\cdot T^2 - 6.024\cdot 10^{-7}\cdot T^3 + 1.286\cdot 10^{-10}\cdot T^4 \quad (22)$$

The thermal conductivity is:

$$k_{air} = 0.00084 + 9.502\cdot 10^{-5}\cdot T - 3.594\cdot 10^{-8}\cdot T^2 - 7.668\cdot 10^{-12}\cdot T^3 \quad (23)$$

Where μ is the dynamic viscosity, c is the thermal capacity at constant pressure, k is the thermal conductivity and T is the fluid temperature in Kelvin. The air was treated as an ideal gas when computing the density.

3.3.8. Material physical properties

Because of the high working temperature of the solar absorber, the conductivity of the alloy-310 is considered temperature dependent using the following polynomial function [49]:

$$k_{310} = 4.0106 + 0.02750782\cdot T - 6.173461\cdot 10^{-6}\cdot T^2 \quad (24)$$

Where k_{310} is the 310 alloy thermal conductivity and T is the 310 alloy temperature in Kelvin.

3.4. Numerical method

Various configurations with dense wire mesh absorber are studied using a commercially available CFD code, STAR-CCM+ with user defined functions, using a three-dimensional model to study a non-homogeneous incident concentrated solar flux (Fig. 4). The physical model consists on a simple cylindrical pipe with different thicknesses depending on the absorber analyzed (Table 1). The user functions are coupled with STAR-CCM+ solver for enhancing the standard features [50]. The user defined functions are used to define boundary conditions, material and fluid properties and source terms. For the LTNE model, the additional scalar equation and source terms in the solid energy equations are solved through passive scalars and user-defined-functions. The velocity and the pressure coupling are handled with SIMPLE algorithm. The momentum equation, the energy equations and P_1-approximation are discretized by adopting the second-order upwind model. The set of governing equations is solved in a segregated way, which means that the discretized momentum, energy and passive scalar equations are solved one by one during the iterations. The convergence criterion was 10^{-5} for all the simulations.

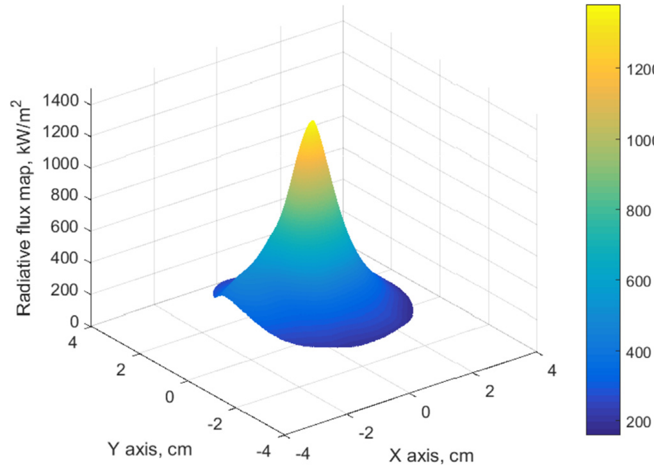


Fig. 4. Representative experimental concentrated flux profile.

4. Results and discussion

4.1. Model comparison with experiments

This section presents a comparison of numerical model results with experimental data available in the literature for dense wire mesh stack under non-homogeneous concentrated radiation [24]. The numerical results were obtained assuming a perfectly insulated absorber sample which is close to past experiments conducted with these meshes [24].

The experimental tests were performed with similar incident radiation distribution with average values between 400 and 420 kW/m^2. Four different volumetric flow rates (7 m^3/h; 6 m^3/h; 5 m^3/h; 4 m^3/h) were set for each wire mesh screen and each test was repeated at least three times. For the numerical simulation a representative incident flux profile was considered (Fig. 4). The HEM using LTNE model coupled to the P_1-approximation was adapted to the experimental working conditions, in terms of incident flux and inlet fluid velocity. Moreover, four simulations (corresponding to the four different mass flow rates) were performed for each mesh type (A; B; C; D; E and F). The average values of the experimental results for each mesh are considered and compared with the simulation results. These results are the average temperature of the fluid at the absorber outlet ($T_{f\text{-}out}$), the thermal efficiency of the absorber (η_{abs}) and the front temperature of the solid matrix ($T_{s\text{-}in}$).

4.1.1. Average temperature of the fluid at the absorber outlet

Table 5 presents the experimental (E) and numerical (N) results of the average fluid outlet temperature and absorber thermal efficiency for the six OVR with the deviation (Dv). The comparison shows a good agreement between the experimental and numerical average fluid temperatures at the absorber outlet. Four absorbers (A; B; C; E) presented temperature deviations lower than 3% and the remaining two configurations (D and F) presented a maximum deviation of 6.4%. The discrepancies found for these last two meshes show the numerical simulation underestimate the average outlet air temperature and thus the absorber efficiency. The D and F meshes have large specific surfaces allowing the fluid to reach quickly high temperatures.

4.1.2. Absorber thermal efficiency

Considering the absorber thermal efficiency (η_{abs}) the maximum relative difference (deviation) between the

Table 5
Experimental and numerical results for 6 single porosity dense wire mesh OVR tested at four different volumetric flow rates.

Volumetric flow rate		7 m^3/h			6 m^3/h			5 m^3/h			4 m^3/h		
Mesh	Variable	E	N	Dv	E	N	Dv	E	N	Dv	E	N	Dv
A	$T_{f-out}(K)$	587	590	0.6	628	632	0.7	681	683	0.3	753	752	−0.1
	$\eta_{abs}(\%)$	85.1	86.6	1.6	83.5	84.2	0.9	81.3	81.4	0.1	77.8	77.4	−0.6
B	$T_{f-out}(K)$	590	601	1.9	630	645	2.3	681	699	2.6	749	772	3.0
	$\eta_{abs}(\%)$	86.9	89.9	3.3	84.8	87.7	3.3	82.3	84.9	3.1	77.9	80.9	3.7
C	$T_{f-out}(K)$	595	602	1.1	641	647	0.9	693	701	1.1	764	774	1.4
	$\eta_{abs}(\%)$	89.2	90.1	0.9	87.8	88.0	0.3	84.7	85.3	0.7	80.3	81.3	1.2
D	$T_{f-out}(K)$	578	564	−2.5	619	600	−3.2	670	641	−4.5	739	695	−6.4
	$\eta_{abs}(\%)$	84.7	78.6	−7.8	83.2	75.8	−9.8	80.6	72.2	−11.6	75.5	67.2	−12.3
E	$T_{f-\mathbf{out}}(K)$	589	601	2.1	629	646	2.6	683	701	2.6	751	775	3.1
	$\eta_{abs}(\%)$	86.8	89.8	3.4	85.1	87.9	3.2	82.5	85.3	3.3	78.3	81.5	4.0
F	$T_{f-out}(K)$	568	543	−4.6	609	577	−5.6	645	617	−4.5	–	–	–
	$\eta_{abs}(\%)$	81.0	72.4	−12.0	79.4	70.0	−13.5	74.3	67.1	−10.8	–	–	–

E (experimental); N (numerical); Dv (numerical-experimental relative deviation).

experimental and numerical efficiency for configurations with mesh type A, B, C and E is lower than 4% and for the absorbers with meshes type D and F the maximum deviation is 13.5% (Table 5).

4.1.3. Solid matrix frontal temperature

Regarding the measurement of the frontal temperatures of the solid matrix, it is a complex procedure as it depends on several factors: incident flux profile, absorber emissivity, ambient temperature, distance from the infrared camera to the sample, etc. Fig. 5 to Fig. 8 shows the experimental measurements of the frontal solid temperature for meshes type A, B, C and E respectively. The main results of the measurements are:

- The frontal solid temperature of meshes type A and B obtained experimentally could not be compared with the numerical data given that there was great variability in the frontal temperature (Figs. 5b and 6b). It is observed that the mesh number (M) has an important effect over the measurement of the frontal temperature: the lower value of the M parameter is (Table 1), the higher the free distance between two consecutive wires (mesh pitch). That causes that the temperature captured by the infrared camera is a mix between several screens.

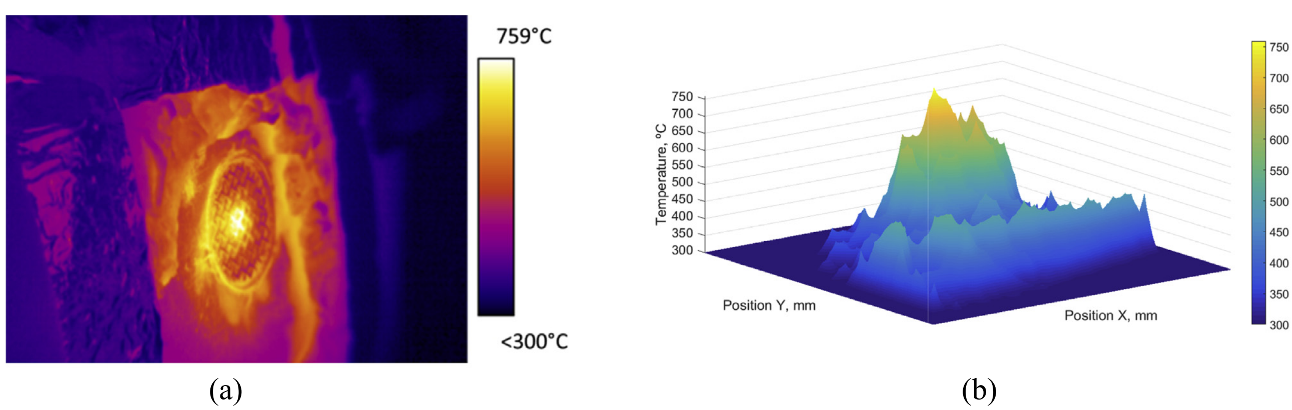


Fig. 5. Mesh type A experimental frontal temperature. (a) Infrared camera. (b) Processed image.

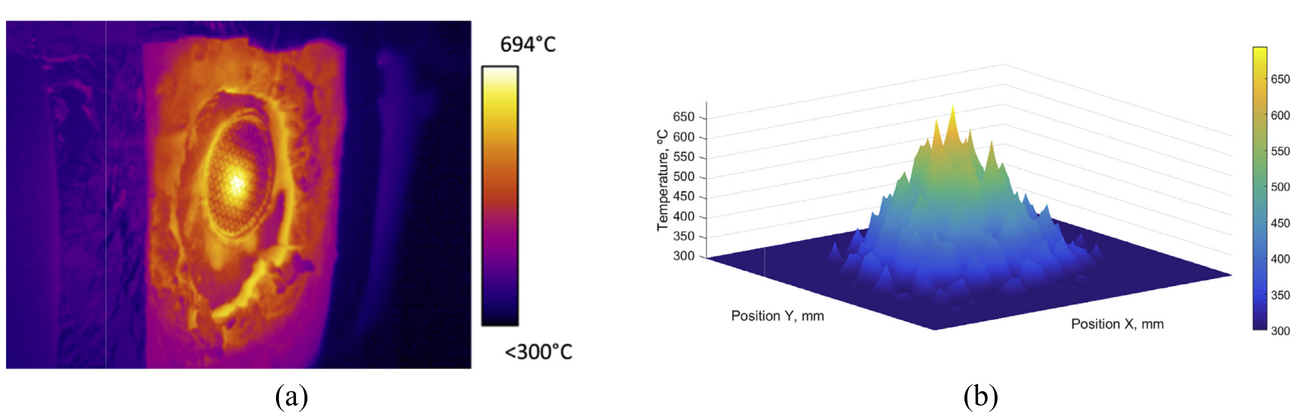


Fig. 6. Mesh type B experimental frontal temperature. (a) Infrared camera. (b) Processed image.

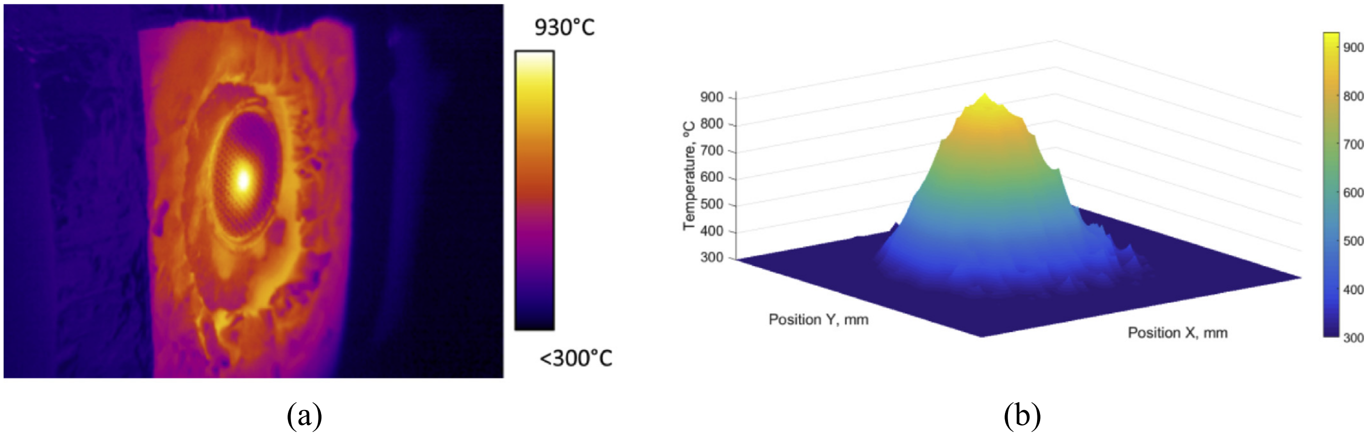


Fig. 7. Mesh type C experimental frontal temperature. (a) Infrared camera. (b) Processed image.

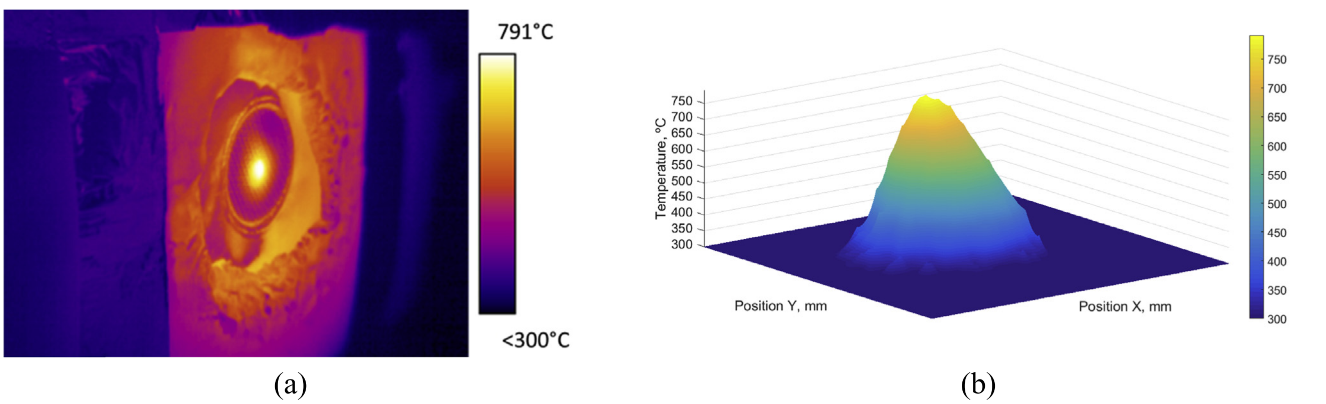


Fig. 8. Mesh type E experimental frontal temperature. (a) Infrared camera. (b) Processed image.

- The frontal solid temperature was only measured reliably on type C (Fig. 7) and E (Fig. 8) meshes as they have a mesh pitch that allows an experimental measurement of just the first screen.
- And last, recording the temperature of type D and type F meshes was difficult given that during the tests it suffered structural deformations constantly that gave erroneous measurements with its changing positions. So these meshes were no further considered for the frontal temperature analysis.

After the previous analysis, the comparison between experimental and numerical results is focused in meshes type C and E for 7 m^3/h volumetric flow rate. The comparison is focused in the solid temperature profiles in two directions: North-South axis (Fig. a) and East-West axis (Fig. b) as the concentrated flux profile is non-homogeneous (Fig. 4).

4.1.4. Absorber with mesh type C

The results show that the maximum deviation usually occurs in the peak of the Gaussian curve. The maximum deviation is −4.7% in the North-South axis, with a predicted temperature in the peak of 1149 K and a measured temperature in the peak of 1203 K (Fig. 9).

The deviation is produced in the North-South axis, which is explained by a slight movement of the peak flux (of the solar simulator) in between 0.5 and 1 mm to the North direction.

4.1.5. Absorber with mesh type E

The maximum deviation is found in the North-South axis and is lower than 0.8%, with a predicted temperature in the peak of 1072 K and a measured temperature of 1064 K (Fig. 10). As well as in the previous absorber, the peak is displaced from the center by 0.5 mm to the North direction.

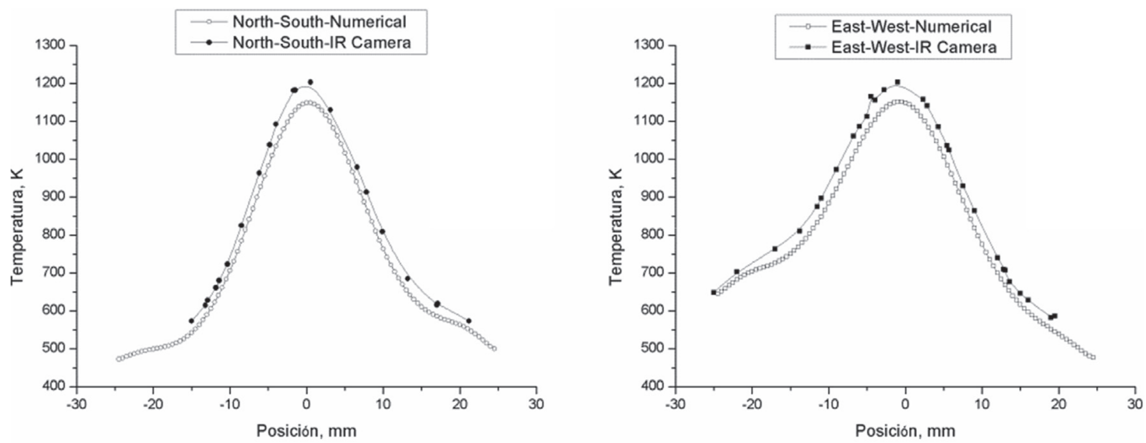
4.1.6. Summary

Based on this comparison between experimental and model results, it is concluded that the HEM and the methodology reproduces well the outlet fluid temperature and absorber efficiency for the meshes A, B, C and E together with the frontal solid temperature for those meshes with an intermediate mesh count (M) value, as happens with meshes type C and E.

However, the discrepancies for meshes D and F are attributed to the radiative heat transfer model. As the HEM tends to overestimate the fluid temperature at the entrance [22], especially for meshes D and F, the solid temperature and the absorber thermal efficiency are lower for the numerical model because the P_1-approximation overestimate the thermal losses.

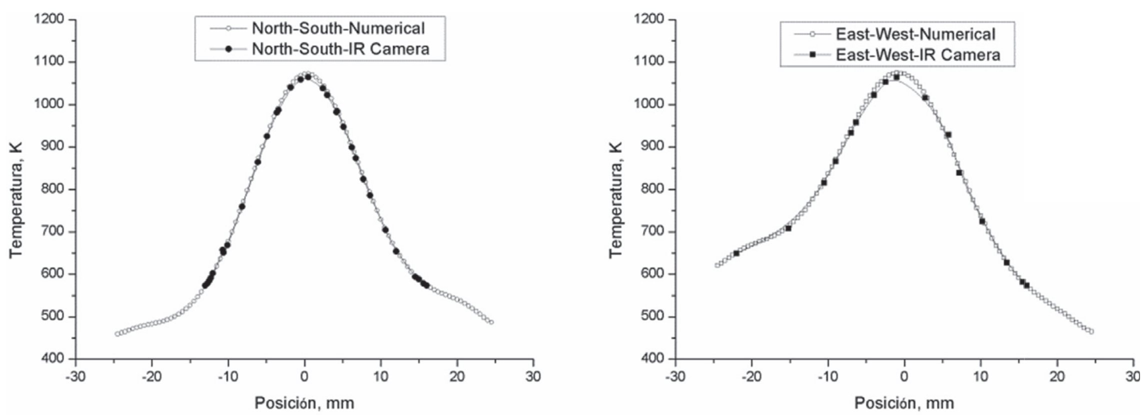
Furthermore, it is noteworthy to mention that for the numerical modeling the emissivity was set constant because it could not be determined precisely so an uncertainty is associated with the solid temperature profiles measurements.

Thus, the numerical model of the volumetric absorber made of dense wire meshes is considered as an acceptable tool to analyze



(a) (b)

Fig. 9. Measured and predicted frontal temperature of the solid matrix for absorber with mesh type C and a volumetric mass flow rate of 7 m^3/h. (a) North-South axis; (b) East-West axis.



(a) (b)

Fig. 10. Measured and predicted frontal temperature of the solid matrix for absorber with mesh type E and a volumetric mass flow rate of 7 m^3/h. (a) North-South axis. (b) East-West axis.

the wire mesh absorber performances.

Concerning with the extrapolation of the results presented in this work to a larger receiver, several considerations have to be considered based on the previous experiences at PSA: (a) a larger receiver will be based on an array of cups, similar to the Hitrec receiver, the Solair receiver or the Jülich power plant [7]; (b) each cup is considered as a quasi-independent sub-receiver, as the volumetric absorber of each cup does not interact in-between; (c) the only interaction between cups would be in terms of radial conduction; (d) the solar flux over each cup could be considered as a nearly-constant incident flux. Obviously, the radiation profile could varies significantly between different cups, but the mass flow rate can be adjust with different methodologies, i.e. the outer cups of the complete receiver can have a higher outlet cup diameter in order to reduce the air velocity, or the cups located at the center can use a lower diameter (the cup diameter is the same, but the isolation thickness could be varied), moreover, perforated plates can be adopted to control the mass flow rate as well; (e) finally, once the air mass flow rate of each cup is known, the efficiency of each cup can be computed independently, and finally the overall receiver efficiency is obtained. In summary, the results presented here could be extrapolated to a larger receiver.

4.2. Thermal study of absorbers with dense wire mesh

A thermal study under fixed and constant input conditions was performed to study the behavior of the volumetric absorbers experimentally tested, isolating the possible effects caused by a heterogeneous flux map. Following the assumptions presented in previous works [24], six absorbers with single porosity are simulated, but with the constraint of having pairs with similar porosities and different geometrical and optical properties. The concentrated solar flux of 600 kW/m^2, air inlet temperature of 300 K and air superficial velocity of 1 m/s applies to all the cases of this section.

The HEM presents a wide variety of accessible quantities that allows deep physical insight into the absorber mechanisms. Exemplary results of the model are displayed in Figs. 11–13.

Fig. 11 shows the solid and fluid temperature profiles for six different meshes and Table 6 presents the most representative geometrical and optical properties, fluid and solid temperatures and absorber thermal efficiency for six absorbers with stagger stack meshes of single porosity. The results show that:

Table 6
Representative geometrical and optical properties, fluid and solid temperatures and absorber thermal efficiency for six stagger stack meshes absorbers with single porosities.

Mesh type	∅	d_h	a_v	β	T_{s-in}	T_{s-out}	T_{f-out}	η_{abs}
	%	mm	m^{-1}	m^{-1}	K	K	K	%
A	70.1	2.35	1194	760	993.4	706.4	705.2	81.9
B	67.6	1.46	1849	1177	897.7	721.1	720.2	85.1
C	62.0	0.81	3044	1938	801.3	725.9	725.8	86.3
D	61.8	0.26	9552	6081	691.9	684.6	684.6	77.7
E	47.7	0.57	3322	2115	764.3	728.5	728.3	86.8
F	46.9	0.12	16330	10396	662.0	657.4	657.3	72.0

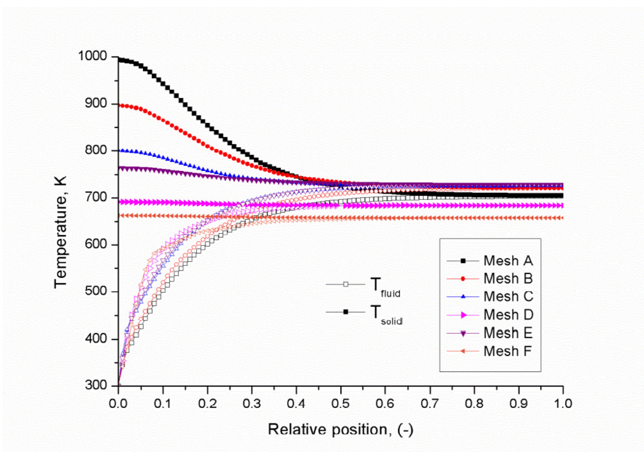


Fig. 11. Solid and fluid temperature profiles for six different dense wire mesh absorbers for 1 m/s (see Table 3).

- The absorbers with mesh type E and C are the ones with the highest efficiencies and air outlet temperatures.
- The absorbers with mesh type F and D are the ones that behave the worst, having the lowest temperatures and efficiencies.
- The other absorbers (mesh type A and B) have an intermediate performance among the six absorbers.

Fig. 12 shows the divergence of the radiative heat flux, included as a source term in equation (4), for the six configurations studied. The results present that the divergence achieve the maximum values for type F absorber due to a higher absorption coefficient (equation (18)), while type A absorber takes the lower divergence value. The results of the divergence of the radiative heat flux match up with extinction coefficient for each absorber type as shown in Table 6. The higher the extinction coefficient is, the lower the penetration of the solar radiation and the higher the absorption near the frontal surface.

Fig. 13 shows the results of the local volumetric HTC for the six absorbers, which is the parameter that couples the fluid energy equation (Eq. (3)) with the solid energy equation (Eq. (4)). As presented in previous works, the HTC is linked with the specific surface area [22], so the higher the specific surface area is, the higher the volumetric HTC. Moreover, it is seen that the HTC is an essential parameter for the determination of the frontal solid temperature, so the higher the local volumetric HTC is, the lower the frontal solid temperature, as shown in Fig. 11.

Overall, the results presented in Figs. 11–13 have the same order of magnitude to that presented by Zaversky et al. [20] for ceramic foams.

As a summary of the analysis, Table 6 presents some of the most important geometrical and optical properties, fluid and solid temperatures and absorbers efficiencies. It is concluded that the absorbers thermal efficiency is mainly affected by two parameters: the volumetric HTC (Fig. 13) and the mean beam length of extinction, 1/β, (Table 6):

- Increasing the volumetric HTC result in a more efficient convective heat transfer between the solid matrix and the fluid.
- Increasing the penetration depth (1/β) produces the absorption of the solar radiation deeper inside the absorber.

Both parameters are related to the porous structure parameters and show conflicting trends (increasing the HTC reduce the penetration depth and conversely). So, from the analysis of the results, in absence of an optimization, it can be concluded:

- The frontal solid temperature is directly related to the specific surface area (which is subsequently linked to the HTC). The lower the specific surface area is, the higher the frontal solid temperature (Table 6).
- The worse thermal performance is produced for those meshes with the highest extinction coefficient (which is subsequently

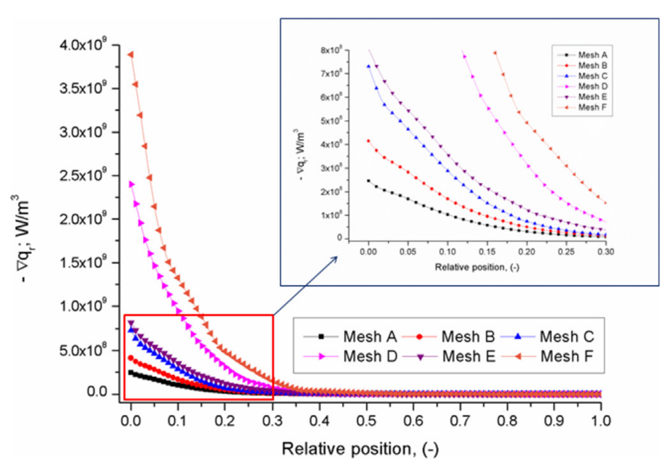


Fig. 12. Results of the divergence of the radiative heat flux as function of the relative absorber depth for six different meshes.

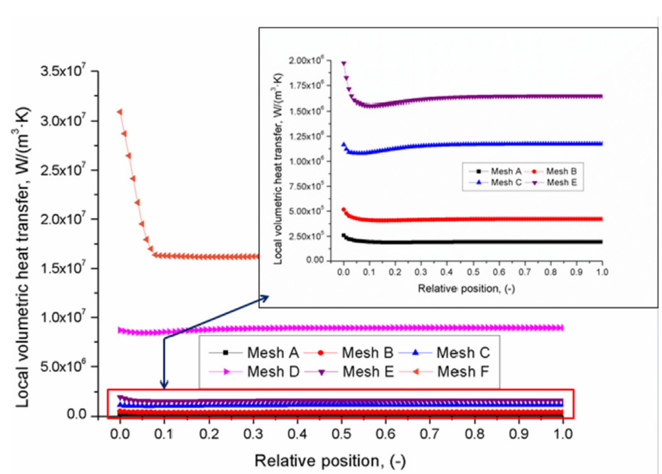


Fig. 13. Results of the local volumetric heat transfer coefficient as function of the relative absorber depth for six different meshes.

linked to the solar transmissivity through the depth of the absorber or the penetration depth), like meshes type D and F, despite having the largest specific surface area (Table 6). Even if the model overestimates the radiative losses (solar reflection and thermal emission) the trend shows a decrease of the thermal efficiency.
- The best thermal performance happens for intermediate properties (specific surface area and extinction coefficient), like meshes type C and E (Table 6).
- The remaining meshes present one good value (medium specific surface area and low extinction coefficient) that give average thermal efficiencies, likes meshes type B and A (respectively) as shown in Table 6.

Finally, according to the numerical results (best behavior for intermediate geometrical and optical properties), the thermal behavior could be further increase performing an optimization analysis. Future works will deal with an optimization analysis of the absorbers behavior with different geometrical and optical properties, with the objective of maximizing the thermal efficiency.

5. Conclusion

This paper presents a numerical model together with the methodology, main assumptions, boundary conditions and parameters needed to simulate, by means of the equivalent porous media approach, the performance of OVR made with dense metallic wire mesh screens. The HEM approach uses the LTNE model together with the P_1-approximation. The OVR adopted a staggered stack arrangement with plain-weave wire mesh screens.

A comprehensive comparison was performed with the experimental data available in the literature. The simulations were performed in a three dimensional geometry due to the non-homogeneous concentrated radiation used in the experimental tests. For comparison between numerical and experimental data the variables adopted were the mean temperature of the fluid at the absorber outlet, the absorber thermal efficiency and the front temperature of the solid matrix. The comparisons show the numerical model presents an acceptable accuracy and the discrepancies are attributed to an overestimation of the radiative thermal losses by the P_1-model and an imprecise determination of the thermal emissivity. The second part of the study focused on the thermal study of different absorbers at a constant concentrated solar flux and volumetric flow rate. The main results are:

- The best thermal performance happens for intermediate properties (specific surface area and extinction coefficient) independently of the porosity.
- The worse thermal performance is produced for those meshes with the highest extinction coefficient (linked to a poor solar transmissivity through the depth of the absorber), despite having larger specific surface area.
- The frontal solid temperature is directly related to the specific surface area (which is subsequently linked to the HTC). The lower the specific surface area is, the higher the frontal solid temperature.
- The air outlet temperature of the fluid is directly linked to the extinction coefficient (if the value is very high), producing a very poor heat transfer transmission.

Future works will deal with a numerical optimization process for dense wire mesh absorber based on the results presented in this paper (best behavior achieved for intermediate geometrical and optical properties).

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Nomenclature

a	Constant value (−)
b	Reynolds constant value (−)
c	Prandlt constant value (−)
C_f	Compactness factor (−)
C	Characteristic mesh constant (−)
d	Diameter (m)
d	Cell size (m)
d_h	Hydraulic diameter (m)
G	Irradiance (W/m^2)
G_s	Diffuse irradiance (W/m^2)
G_c	Collimated irradiance (W/m^2)
h_v	Volumetric heat transfer coefficient ($W/(m^3 \cdot K)$)
I_0	Incident radiation (W/m^2)
i_b	Black body intensity (W/m^2)
K_1	Inertial permeability coefficient (m^2)
K_2	Viscous permeability coefficient (m)
L	Absorber thickness (m)
M	Mesh size (1/m)
P	Pressure (Pa)
q_r	Radiative flux (W/m^2)
T	Temperature (K)
z	Fluid direction (m)
Nu	Nusselt number (−)
Re	Reynolds number (−)
Pr	Prandtl number (−)
η_{abs}	Absorber efficiency (−)
α	Absorptivity (−)
a	Absorption coefficient ($1/m$)
σ	Stefan Boltzmann constant ($W/(m^2 \cdot K^4)$)
σ_s	Scattering coefficient (1/m)
β	Extinction coefficient (1/m)
a_v	Specific surface area (1/m)
ρ	Density (kg/m^3)
k	Conductivity ($W/(m \cdot K)$)
c	Heat capacity ($J/(kg \cdot K)$)
v	Velocity (m/s)
v_D	Darcy velocity (m/s)
$\varnothing$	Porosity (−)
μ	Viscosity ($Pa \cdot s$)

Abbreviations

E	Experimental
N	Numerical
na	Not available
Dv	Deviation
HTC	Heat transfer coefficient
HEM	Homogeneous equivalent model
IR	Infrared

Subscripts

310	310 alloy
amb	Ambient

c	Collimated
eff	Effective
f	Fluid
in	Inlet
l	Local
v	Volumetric
out	Outlet
s	Solid

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